Chemical Names and Formulas

Answer key with worked solutions

A. Ionic or molecular?

1. Classify.

> (a) **ionic** (metal K) (b) **molecular** (non-metals only) (c) **ionic** (ammonium ion) (d) **molecular** (e) **ionic** (metal Fe)

B. Ionic compounds

2. Formulas.

> (a) K^+, O^{2-}: $2(+1) + (-2) = 0$, so $\mathbf{K_2O}$
> (b) Mg^{2+}, N^{3-}: $3(+2) + 2(-3) = 0$, so $\mathbf{Mg_3N_2}$
> (c) Zn^{2+}, Cl^-: $(+2) + 2(-1) = 0$, so $\mathbf{ZnCl_2}$
> (d) Ba^{2+}, $SO_4{}^{2-}$: $(+2) + (-2) = 0$, so $\mathbf{BaSO_4}$

3. Formulas with polyatomic ions.

> (a) Ca^{2+}, $NO_3{}^-$: $(+2) + 2(-1) = 0$, so $\mathbf{Ca(NO_3)_2}$
> (b) Al^{3+}, OH^-: $(+3) + 3(-1) = 0$, so $\mathbf{Al(OH)_3}$
> (c) $NH_4{}^+$, $CO_3{}^{2-}$: $2(+1) + (-2) = 0$, so $\mathbf{(NH_4)_2CO_3}$
> (d) Na^+, $HCO_3{}^-$: $(+1) + (-1) = 0$, so $\mathbf{NaHCO_3}$

4. Names.

> (a) **sodium sulfide** (b) **sodium sulfate** (c) **silver nitrate** (d) **magnesium phosphate**

5. Roman numerals.

> (a) 2 Cl^- give −2, so Fe is +2: **iron(II) chloride**
> (b) 3 O^{2-} give −6, shared by 2 Fe, so each Fe is +3: **iron(III) oxide**
> (c) 1 O^{2-} gives −2, shared by 2 Cu, so each Cu is +1: **copper(I) oxide**
> (d) 1 $SO_4{}^{2-}$ gives −2, so Cu is +2: **copper(II) sulfate**

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C. Molecular compounds

6. Prefixes.

> (a) **carbon dioxide** (b) **phosphorus pentachloride** (c) **dinitrogen monoxide** (nitrous oxide) (d) **sulfur hexafluoride** (e) SO_3 (f) N_2O_4

D. Acids

7. Acids.

> (a) HCl (b) HNO_3 (c) H_2SO_4 (d) H_3PO_4

E. Challenge

8. Spot the errors.

> - Copper forms Cu^{+} and Cu^{2+}, so the name needs a Roman numeral. In copper(II) sulfate, $(+2) + (-2) = 0$, so one sulfate is enough: **copper(II) sulfate is** $CuSO_4$. $Cu(SO_4)_2$ would need copper to be +4.
> - $CaCl_2$ is ionic, so prefixes are not used, and the formula contains one calcium, not two: **calcium chloride**.

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